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MECHATRONICS ENGINEERING JUNIOR

ARNAV THOMAS

I love turning digital concepts and simulations into working physical hardware.
My core skills:

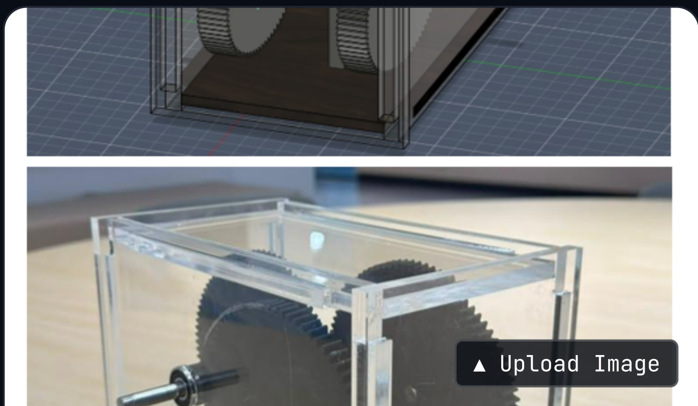
1) Fusion 360 and AutoCAD for 3D modeling, gear assemblies, and housing layouts.

2) MATLAB for data cleaning, regression modeling, and statistical validation.

3) C programming, Webots robotics, state machines, and analog circuits.

4)Rapid prototyping using 3D printing and laser-cut acrylic.

MECHANICAL DESIGN & FABRICATION



GEAR TRAIN CAD & MODELING

Miniature Catapult

Designed and manufactured a high-repeatability miniature catapult constrained to a 200 x 200 x 200 mm bounding box.

Inspect Technical Architecture ➔

Model Winch for Foundry

Designed a 3-stage compound spur gearbox to lift a 1 kg crucible load vertically within a compact footprint under a 250 AED budget limit.

Inspect Technical Architecture ➔

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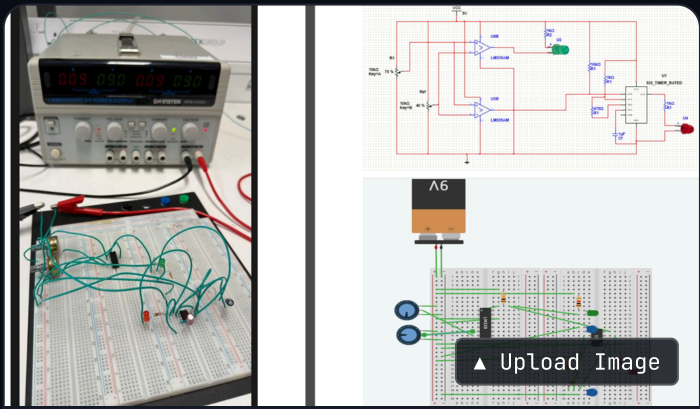
ROBOTICS SIMULATION & CONTROL

Path-Clearing Robot

Programmed and simulated an autonomous path-clearing robot in Webots using a robust state-machine loop in C.

Inspect Technical Architecture ➔



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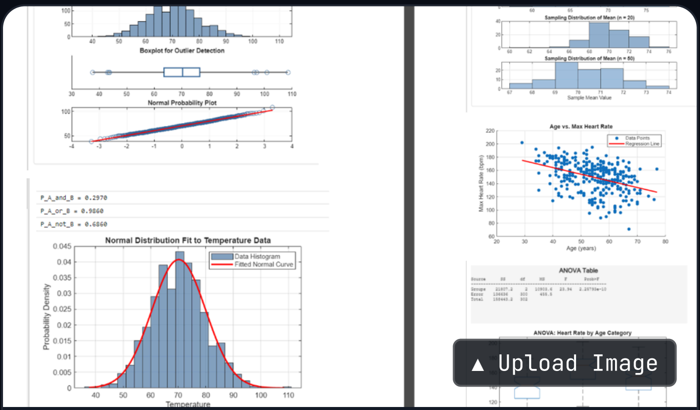
ANALOG ELECTRONICS DESIGN

Temperature Threshold Alarm

Designed and simulated an analog thermal monitoring and alert circuit featuring dual status notifications.

Inspect Technical Architecture ➔



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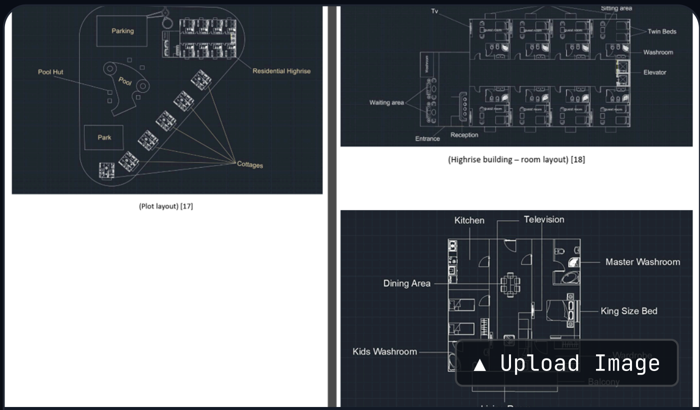
MACHINE DIAGNOSTICS & DATA

Industrial Machine Health Modeling



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INFRASTRUCTURE PLANNING & MODELING

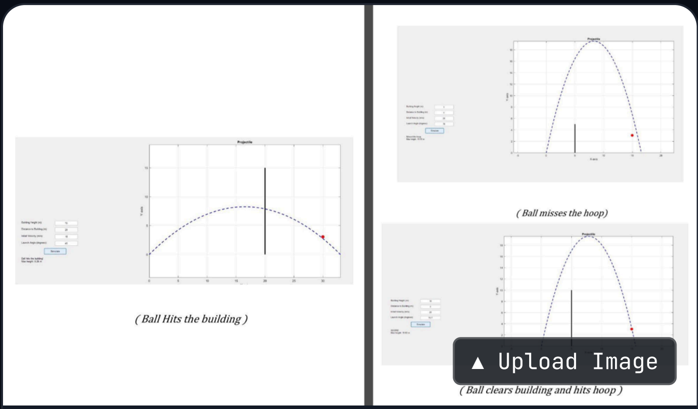
Hybrid Microgrid Energy System Design

Designed and implemented an end-to-end data processing and predictive health model in MATLAB for 1,000 sensor logs.

Inspect Technical Architecture ➔

Modelled and optimized a hybrid, self-sustaining renewable microgrid to support a proposed 2,000 sqm development in Hanoi.

Inspect Technical Architecture ➔



MATLAB PROGRAMMING

Projectile Simulation Project

Built a neat 3D math model and MATLAB simulator to see if we could launch a ball over a building and land it perfectly on a target

Inspect Technical Architecture ➔